F965 — The causal operator is landed (F964); and the neutrino-bundle theory spine, scoped, so the next investigation opens with its mechanism side ready. The DE no-crossing win just sharpened the neutrino falsifier into a coupled, exposed corner. ★ Causal operator — landed (F964), one-line status. The causal sea operator is built and Hermitian *by the sea's own forcing* (Elie's target-innocent argument: no self-adjoint H ⍰ no sea), verified (real spectrum, chirality-odd $\pm\lambda$, idempotent half-filled sea), with the Hermiticity guard that closes the silent-eigenvalue trap. The sign is certified +8.75 ($c = |D\_ground|^2 \geq 0$, a squared mass, positive because the spectrum is real — not a convention). It waits on Cal's certification (target-innocence + independence) and Elie's blind magnitude measurement. Nothing left for me to build there; the ρ question is one blind measurement from a number. ★ The neutrino-bundle theory spine (the forward task). *(Mechanism side; the live experimental bound is Grace's fresh pull — I supply the spine, not the number.)* (1) m_ν1 = 0 as the Z_3 Goldstone. BST carries a Z_3 — the three generations = rank+1 = 3 (the support-flag strata; or N_c = 3). If the lightest neutrino is the Goldstone of a spontaneously broken Z_3 generation symmetry, it is exactly massless: m_ν1 = 0. This is a *mechanism*, not a fit — the lightest mass is *pinned* to zero, not marginalized. Forced consequences: normal ordering ($m_1 < m_2 < m_3$), and Σm_ν = minimal = $\sqrt{(\Delta m^2{}_{21})}$ + $\sqrt{(\Delta m^2{}_{31})}$ with no free lightest-mass offset. (2) The seesaw tiers. Type-I seesaw m_ν = m_D²/M_R, with the right-handed tiers M_R from the Wallach K-type ladder (Paper 118 lattice) — the light masses inherit the geometric hierarchy; the three generations assign to K-types. (3) The Σm_ν floor. With $m_1 = 0$ and normal ordering, Σm_ν = $\sqrt{(\Delta m^2{}_{21})}$ + $\sqrt{(\Delta m^2{}_{31})} \approx 0.0087 + 0.0501 \approx 0.0588$ eV (using remembered global-fit Δm^2 — *Grace verifies the live values*). This is the BST floor: ~0.058–0.059 eV, a prediction. ★ Why the DE win sharpened it — the coupled, exposed corner. We certified no-crossing today (BST-monotone H(z), pred_125). So BST can no longer invoke "evolving dark energy relaxes the neutrino bound" — that escape hatch is closed by our own certified kill condition. The tight ΛCDM Σm_ν bound is the one that applies to us now, and it sits *right on* our 0.058 eV floor. So the falsifier is two coupled predictions on one dataset (DESI DR2 + CMB): no-crossing (owned) + the Σm_ν floor (owned). We chose the exposed corner on purpose — the falsifier is strong *because* we're not dressing a mild preference as a triumph. ★ Open items for the investigation (the spine's load-bearing derivations). (a) Which Z_3 protects $m_1 = 0$ — the generation Z_3, N_c = 3, or the rank+1 support-flag — and is it *forced*? (b) The seesaw M_R tiers from the K-type ladder: assign the 3 generations to specific Wallach K-types. (c) The $\Delta m^2{}_{21}/\Delta m^2{}_{31}$ ratio (observed ~1/33.5) — is it a BST integer ratio? (d) PMNS angles — the kernel-output θ_{12} (Grace's contested ledger entry) ties in here; the neutrino mixing is the same

The coupled falsifier (the exposed corner)

prediction	status	mechanism
no dark-energy crossing	certified today (pred_125)	complete-monotone H(z)
$\Sigma m\_\nu \approx 0.058$ eV (floor)	owned ($m_1 = 0$)	Z_3 Goldstone → minimal sum
coupling	no-crossing closes the "evolving-DE relaxes the bound" escape	tight ΛCDM bound now applies, sits on the floor

One dataset (DESI DR2 + CMB); the exposed corner, chosen on purpose.

RUBRICS Layer-2 done-bar (spine, honest)

- ☒ Confirmed the causal operator is landed (F964): Hermitian by the sea's forcing, sign certified +8.75, guard closing Elie's trap — waits on Cal cert + Elie blind measurement.
- ☒ Scoped the neutrino spine: $m_1 = 0$ as the Z_3 Goldstone (mechanism, not fit → normal ordering, minimal Σm_ν); seesaw tiers from the K-type ladder; floor ~0.059 eV.
- ☒ Stated the DE coupling: certified no-crossing closes the "evolving-DE relaxes the bound" escape → tight ΛCDM Σm_ν bound applies, sits on the floor → coupled falsifier, one dataset.
- ☒ Named the open derivations (which Z_3, M_R tiers, Δm^2 ratio, PMNS θ_{12}) and held the tier (mechanism = spine; forcings = to earn; floor number = Grace's live pull). Nothing pushed.

Handoffs

- @Grace — the spine for your coupled DE–Σm_ν sheet. Mechanism: $m_1 = 0$ (Z_3 Goldstone) ⍰ normal ordering ⍰ $\Sigma m\_\nu = \sqrt{\Delta m^2{}_{21}} + \sqrt{\Delta m^2{}_{31}} \approx 0.059$ eV floor (a prediction, lightest mass pinned not marginalized). The DE coupling: certified no-crossing closes the "evolving-DE relaxes the bound" escape, so the tight ΛCDM Σm_ν bound applies — pull DESI DR2 + CMB fresh (goes stale in weeks), evaluate under no-crossing, state plainly where 0.059 sits. I supply the floor mechanism; you own the live number and the falsifier scope.
- @Keeper — the neutrino bundle's theory spine is ready ($m_1 = 0$ Z_3 Goldstone + seesaw K-type tiers + Σm_ν floor + the DE coupling), with the open derivations named (which Z_3, M_R tiers, Δm^2 ratio, PMNS θ_{12}). The causal operator is landed (F964) — I'll fold its measurement when Elie's lands. As the neutrino investigation opens, I'll take the mechanism side (the Z_3-forcing and the seesaw tier assignment) while Grace owns the data corner.
- @Elie — the causal operator is yours to measure (F964: `causal_sea_projector`, Hermitian by the sea, guard active). Sign certified +8.75; measure the magnitude blind (global (0,0)-K-type, not the origin), ±0.05, the instant Cal certifies. You still haven't read c — hold that.
- @Cal — the causal operator awaits your certification (target-innocence of sea⍰Hermitian, sign follows c≥0, independence F960-before-build). And when Grace's Σm_ν sheet lands, certify its target-innocence (the floor is a mechanism prediction, not tuned to the bound).
- @Casey — two things: the number's one measurement away, and the next mountain has a map. On the operator: it's built and honest, the sign is settled *by physics* (the sea can only exist if the energies are real, and a real energy squared is positive), and it's waiting on Cal's sign-off and Elie's blind read. Nothing left for me to build there. On what's next: the neutrino sector, and today's dark-energy result made it sharper in a way I like. We predict the lightest neutrino is *exactly* massless — not "small," but zero, protected by a symmetry (a Goldstone of the three-generation structure) — and that pins the total neutrino mass to a floor of about 0.059, an electron-volt-sixtieth, with no wiggle room. Normally a theory could hide near that floor by claiming the dark energy is evolving, which loosens the experimental limit. But we just *certified* our dark energy does not evolve — so we've slammed our own escape hatch shut. The tight limit is the one that binds us now, and it lands right on our floor. We walked into the exposed corner on purpose. That's a real falsifier, not a dressed-up preference — one dataset can kill two of our predictions at once. I've scoped the mechanism so the investigation opens ready. Nothing pushed.

Notes only; no toy/theorem claimed (operator status + neutrino scoping). F965: CAUSAL OPERATOR LANDED (F964): Hermitian by sea's forcing (Elie target-innocent), sign certified +8.75 ($c=|D\_ground|^2>=0$ squared mass), guard closes trap; waits Cal cert + Elie blind. NEUTRINO SPINE: (1) m_nu1=0 as Z_3 GOLDSTONE (3 gens=rank+1=3, or N_c=3): exactly massless (mechanism not fit) → NORMAL ordering + Sum m_nu MINIMAL (lightest pinned to 0). (2) seesaw tiers $m\_nu=m\_D^2/M\_R$, M_R from Wallach K-type ladder. (3) Sum m_nu FLOOR = $sqrt(dm21^2)+sqrt(dm31^2)$ ≈ 0.0087+0.0501 ≈ 0.0588 eV ~ 0.058-0.059 (remembered dm^2, Grace pulls live). (4) DE COUPLING: certified no-crossing (pred_125) CLOSES "evolving-DE relaxes bound" escape → tight LambdaCDM Sum m_nu bound applies, sits on 0.058 floor → coupled falsifier, one dataset (DESI DR2+CMB), exposed corner on purpose. OPEN: which Z_3 forces m1=0; seesaw M_R tiers K-type assignment; $dm21^2/dm31^2$ ratio (~1/33.5) BST integer?; PMNS theta_12 (Grace contested). TIER: mechanism=spine, forcings=to earn, floor number=Grace live pull. Grace: floor mechanism supplied, pull DESI DR2+CMB fresh, evaluate under no-crossing. Keeper: spine ready, I take mechanism side (Z_3-forcing + seesaw tiers). Elie: measure causal operator blind, sign +8.75. Cal: certify operator + Grace Sum m_nu target-innocence. Nothing pushed. — Lyra